CISC 3667: Game Design and Development

Homework 2: Randomness and Probability

Assigned: Tuesday, September 29
Due: Thursday, October 1 **before class**
Total Points Available: 50

Show your work for all problems. Answers without supporting calculations may receive partial credit.

1. A die is rolled, and a coin is tossed. Find the probability that **the die shows an odd number** and **the coin shows a head**. (3 pts)

2. Assume that you pick **two cards from a deck of cards** (without replacing the first card before you pick the second). What is the probability that **both are number cards** (i.e., numbers 2 -- 10, not J, Q, K, A)? (4 pts)

3. In the game of Tetris, there are seven shapes (called tetrominoes) that descend during the game. The choice of which tetromino to select next is random, but there is more than one way to implement this randomness.

   Here are two possible random selection algorithms:

   > ***Random A****:* At every iteration, **select a tetromino at random from all seven possibilities.**
   >
   > ***Random B****:* At the beginning of the game, make a "bag" of seven tetrominoes, one of each shape. At every iteration, **choose a tetromino at random from the bag**. **After the tetromino is used, it is discarded from the bag. After seven iterations, create a new bag and proceed as above.**

   a. Suppose you have just been given a certain tetromino. What is the probability that the next piece will have the same shape, using the Random A strategy? (4 pts)

   b. Suppose you have just been given a certain tetromino. What is the probability that the next piece will have the same shape, using the Random B strategy? (4 pts)

4. I'm working on a new tabletop RPG / strategy game / party game. It's pretty promising, but I'm a little over my head on some of the math. So I'm glad I just hired you on as a designer!

   Okay, so think I have a new combat mechanic worked out, but I'm not sure. I need to know the following probabilities to be sure. Can you tell me the following? Make sure to show your work so I can explain it to our publishers later. (I'll use this shorthand a lot, so just FYI: "2d6" means rolling two six-sided dice at once. So "4d10" means four ten-sided dice, and so on.)

   a. Rolling **2d4**, what is the probability of **rolling at least one 4**? (3 pts)

   b. Rolling **2d6**, what is the probability of **rolling at least one 6**? (3 pts)

   c. Rolling **2d6**, what is the probability of **getting both 5s**? (3 pts)

5. I think we need to simplify our combat system.

   I'm thinking it'll work like this:

   - **First, the player rolls a d4**; if they **roll a 4,** they **roll a d6**. If they **roll a 6 on that, then they roll a d20**.
   - If they **roll higher than 15 on that, then they win!**

   I think all the rolling will be pretty exciting.

   What percentage of the time will a player win? (6 pts)

6. The working title of our game is "The Six Trials of the Hero".

   Right now, the core mechanic is pretty simple: the player **rolls a D10 each turn**, and they **need to roll higher than the turn number to proceed**.

   So they need to **roll a 2 or more on turn 1, a 3 or more on turn 2, and so on**.

   If they get past all six trials, they win the game and become a hero.

   But is it too hard, though?

   What percentage of the time will a player win? (6 pts)

7. Mary is playing an adventure game. She is wandering around her game world, and she comes to a fork in the road.

   The fork has 3 choices:

   - The **left** fork leads to the cave of the **evil ogre**, next to which is a **medium-sized bag of money**.
   - The **middle** fork leads to a **small bag of money**.
   - The **right** fork leads to a **large bag of money**, which is guarded by an **angry dragon**.

   Mary's goal in the game is to collect as much money as she can, without getting killed by the evil ogre or the angry dragon.

   She knows that the **evil ogre is lazy and sleeps 75% of the time** (she can grab the nearby money bag while he sleeps).

   She also knows that the **angry dragon is full of energy and only sleeps 10% of the time.**

   Mary needs to decide what to do; i.e., which fork to take.

   - The **large bag of money** is worth **10 points** in the game
   - The **medium bag** is worth **5 points**
   - The **small bag** is worth **1 point**
   - **Getting killed** is worth **-10 points** (and **if Mary goes near the ogre or the dragon when they are awake, she will definitely be killed**).

   A. What is the expected value of going down the **left** path? (3.5 pts)

B. What is the expected value of going down the **middle** path? (3.5 pts)

C. What is the expected value of going down the **right** path? (3.5 pts)

D. Which way should Mary go? (3.5 pts)